Fiscal Year:	FY 2023	**Task Last Updated:**	FY 11/16/2023
PI Name:	Lawler, John Ph.D.		
Project Title:	Upstream Regulation of Nox2 and Skeletal Muscle Atrophy During Microgravity and Countermeasure Development		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	979-862-2038
Organization Name:	Texas A&M University		
PI Address 1:	Health & Kinesiology		
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PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	06/01/2019	**End Date:**	05/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	11	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 11/14/23). NOTE: End date changed to 05/31/2023 per F. Hernandez/ARC (Ed., 6/20/22). NOTE: Start/end dates changes to 6/1/2019-7/31/2022 (originally 3/1/2019-5/31/2022) per NSSC award documents per F. Hernandez/ARC (Ed., 12/3/2020)		
Key Personnel Changes/Previous PI:	August 2020 report: Dr. Khaled Kamal - Assistant Research Scientist ; Amin Mohajeri - PhD student; Mariam Othman - PhD student (add) Adding Joo Kim (PhD student) Jordyn Johnson - MS student. (remove - left because of COVID) Add: Aggie Research Scholars: Devon Roeming, Samhitha Harvey, Danielle DeCastro, Gracie Barrow, Yasmin Bagheri Victoria Frank, Anish Kumaran, Mickael Gergheis, Kyle Hawley, Fernanda Sifuentes (Aggie Research Scholar) 2022- Graduated: Myles McFarland (Aggie Research Scholar) 2020- ; Mollie Linder (Aggie Research Scholar) 2020- ; Hallie Harris (Aggie Research Scholar) 2020 - ; Francisco Melesio (Aggie Research Scholar) 2020 - ; Sonny Rodriguez (Aggie Research Scholar) 2020- ; Mia Ngyuen (Aggie Research Scholar) 2020- ; Lorrie Hill 2018-2020 (Kinesiology Research Scholar).		
COI Name (Institution):	Fluckey, James Ph.D. (Texas A & M, College Station)		
Grant/Contract No.:	80NSSC19K0432		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Skeletal muscles are dynamic mechanical and metabolic machines that drive body movement and energy expenditure. Skeletal muscles are dynamic tissues that can adapt quickly to alterations in mechanical unloading by altering their mass and muscle fiber cross-sectional area. Skeletal muscle strength and endurance are essential to the health, well-being, and performance of astronauts during spaceflight and upon return to a gravitational environment. The mechanical unloading due to the microgravity (µG) of spaceflight causes muscle fiber atrophy and fiber-type shift of postural muscles in the lower extremities and flexors in the upper extremities. Microgravity also increases the risk of skeletal muscle damage, weakness, and thus the risk of injury upon reloading (e.g., extravehicular activities, Mars). Mechanotransduction is the ability to sense and regulate adaptive responses to increased or decreased loading. New paradigms have emerged from ground analogs of µG that have contributed to a leap of our understanding of mechanotransduction and muscle atrophy. Specifically, the mu-splice variant of neuronal nitric oxide synthase (nNOSµ) was discovered as causal in both muscle hypertrophy with overloading and atrophy with unloading. Our laboratory has found that reactive oxygen species (ROS) directly contribute to both muscle fiber atrophy and fiber-type shift from slow to fast. Pilot Data and cutting-edge research have identified mitochondria, the Nox2 isoform of NADPH oxidase, and upstream angiotensin II receptor 1 (AT1R) as sources of ROS during mechanical unloading. Preliminary Data show that inhibition of Nox2 translocation of nNOSµ away from the sarcolemma, muscle fiber atrophy, and fiber-type shift. However, the upstream mechanisms that regulate Nox2 during µG are poorly understood, impeding progress in space biology and novel countermeasure development. The lack of such knowledge impedes our development in understanding the mechanisms that underlie redox regulation of mechanotransduction in skeletal muscle. This grant application serves as a renewal and extension and Renewal of our research team's NNX13AE45G award, particularly stretching our horizons in understanding how Nox2 assembly is enhanced during microgravity in skeletal muscle. New studies have identified novel inhibitors for proteins recently govern assembly of the Nox2 complex at the cell membrane acid—sphingomyelinase (ASMase) and cyclophilin A. We hypothesize that the following novel countermeasures will protect against nNOSµ translocation and the spaceflight phenotype— (a) the ASMase inhibitor etidronate (Didronel) and (b) cyclophilin A inhibitor TMN-355. We further postulate that Nox2 is causal in ROS-induced suppression of anabolic signaling. The efficacy and specificity of the above countermeasures will be confirmed with gene knockdown experiments. Texas A&M is a rich research environment for NASA research, including the Space Life Science Program. We will use the latest molecular and image analysis tools in the development of highly novel countermeasures against spaceflight sarcopenia during microgravity. Dr. Lawler and Dr. Fluckey's laboratories have continued to be supported by NASA, and are dedicated to finding targeted, antioxidant countermeasures against spaceflight sarcopenia. The ground hindlimb unloading model will be used in short and long-term experiments. Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will also directly translate to skeletal muscle wasting in clinical setting on Earth, an important mission of the Space Biology program. For example, hospitalization, particularly in an ICU (intensive care unit) can reduce skeletal muscle mass by 25%. Cast immobilization can decrease affected muscle mass by 30% as well. In addition, mechanical unloading due to disuse (e.g., bedrest) and illness (e.g., sepsis, chronic obstructive pulmonary disease, chronic heart failure) exacerbates atrophy, weakness, and impedes recovery.
Task Progress:	[Ed. Note: For complete citations, see Cumulative Bibliography.] NASA Task Book Space Biology (SB) 2022-2023 bullet points: - We developed a novel Bioreactor to simulate microgravity, radiation exposure, and over/reloading of skeletal muscle cells. We intend to develop co-cultures. Kamal KY, Othman M, Lawler JM. Proof of Concept: Developing a novel bioreactor for skeletal muscle hypertrophy and atrophy by manipulating uniaxial cyclic strain. ASGSR abstract, 2022, Houston, TX. - We developed a new receptor activator of nuclear factor kappa-B ligand (RANKL) knockdown package with adeno-associated virus serotype 9 (AAV9) gene delivery. RANKL is elevated by microgravity, aging, and Duchenne muscular dystrophy in skeletal muscles. Othman, Mariam A, Khaled Y. Kamal, Devon Roeming, Samhitha Ramanuja, John M. Lawler. RANKL protein knockdown applying AAV9/shRNA Systemic Drug Delivery to mitigate spaceflight-induced muscle atrophy, ASGSR Abstract 2022, Houston, TX. Roeming D, Ramanuja S, Kamal KY, Othman M, Lawler JM. AAV9/shRNA Knockdown of RANKL Protein Expression Utilizing Systemic Drug Delivery: Proof of Concept. Student Research Week, Texas A&M University, 2022. - We used a combination of nutritional interventions (fish oil + curcumin) to mitigate skeletal muscle oxidative stress, atrophy, and fibrosis with microgravity. - Also, fish oil + pectin against cardiac fibrosis. Kamal KY, Hord JM, Wu C, Talcott S, Janini Gomes M, Fluckey JF, Ford JF, Nancy D. Turner, Lawler JM. Combination Nutrition Interventions Against Spaceflight Sarcopenia. Human Research Project Meeting. NASA. Galveston, February, 2022. - Partial loading and high energy particle (HZE) radiation resulted in significant damage and markers of fibrosis in skeletal muscle. This paper has been accepted and is in press.

Task Progress:	Wiggs MP, Lee Y, Shimkus KL, O'Reilly CI, Lima F, Macias BR, Shirazi-Fard Y, Greene ES, Hord JM, Braby LA, Carroll CC, Lawler JM, Bloomfield SA, Fluckey JD. Combined effects of heavy ion exposure and simulated lunar gravity on skeletal muscle. Life Sci Space Res. 2023 May;37:39-49. - Involvement of RANKL, which causes bone loss in spaceflight in sarcopenia and inflammatory cell invasion. Lawler JM, Kamal K, Othman M, Lee J. Myeloid specific knockout of the ghrelin receptor mitigates aging-linked elevation of RANKL, Ca2+ overload, and sarcopenia. Pending ASGSR abstract submission. Kamal KY, Lawler JM. Cellular and molecular signaling meet the space environment. International Journal of Molecular Science. 2023 Mar 22;24(6):5955. doi: 10.3390/ijms24065955. - Assistant Research Assistant Dr. Khaled Kamal was awarded a SHINE scholarship for radiation research. NASA SHINE (Space Health Impacts for the NASA Experience) Training Program- Virtual Space Radiation Curriculum (# NNJ22ZSA001L). Dr. Khaled Kamal received this award and participated in the first annual SHINE (Space Health Impacts for the NASA Experience) Space Radiation virtual course. Kim J, M Othman, KY Kamal, JM Lawler. RANKL and Nox2 signaling in Duchenne Muscular Dystrophy models of skeletal muscle. Texas Affiliate – American College of Sports Medicine. Waco, 2023. Othman M, J Kim, KY Kamal, JM Lawler. Role of Ghrelin Receptor in Sarcopenia: Involvement of Redox Signaling and RANKL. Texas Affiliate – American College of Sports Medicine. Waco, 2023. Lawler, JM. Redox Regulation of nNOS and Skeletal Muscle Atrophy During Microgravity. Graduate Faculty of Nutrition Seminar, Texas A&M University, 2022. Lawler JM, Kamal KY, Othman M, Roeming D, Ramanuja S. Redox Regulation of Mechanotransduction in Skeletal Muscle During Spaceflight: Translation to Duchenne Muscular Dystrophy and New Insights, ASGSR abstract, 2022 Houston, TX.

Bibliography Type:	Description: (Last Updated: 11/13/2025)
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Abstracts for Journals and Proceedings	Othman M, Kim J, Kamal KY, Lawler JM. "Role of ghrelin receptor in sarcopenia: Involvement of redox signaling and RANKL. " 2023 TACSM Annual Meeting (Texas Chapter of the American College of Sports Medicine), Waco, TX, February 22-23, 2023. Abstracts. 2023 TACSM Annual Meeting (Texas Chapter of the American College of Sports Medicine), Waco, TX, February 22-23, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Lawler JM, Kamal KY, Othman M, Roeming D, Ramanuja S. "Redox regulation of mechanotransduction in skeletal muscle during spaceflight: Translation to Duchenne Muscular Dystrophy and new insights." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Abstracts for Journals and Proceedings	Kamal KY, Othman M, Lawler JM. "Proof of Concept: Developing a novel bioreactor for skeletal muscle hypertrophy and atrophy by manipulating uniaxial cyclic strain. " 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
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Articles in Peer-reviewed Journals	Wiggs MP, Lee Y, Shimkus KL, O'Reilly CI, Lima F, Macias BR, Shirazi-Fard Y, Greene ES, Hord JM, Braby LA, Carroll CC, Lawler JM, Bloomfield SA, Fluckey JD. "Combined effects of heavy ion exposure and simulated lunar gravity on skeletal muscle." Life Sci Space Res. 2023 May;37:39-49. https://doi.org/10.1016/j.lssr.2023.02.003 ; PMID: 37087178 , May-2023
Articles in Peer-reviewed Journals	Kamal KY, Lawler JM. "Cellular and molecular signaling meet the space environment. " Int J Mol Sci. 2023 Mar 22;24(6):5955. https://doi.org/10.3390/ijms24065955 ; PMID: 36983029; PMCID: PMC10058013 , Mar-2023